BT4222 Final Report

SPOTLIGHT: A News Recommendation Framework for Homepage Slate Optimisation

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Abstract (Standalone page, within 300 words)

- **Business Objective & RecSys Formulation:** Define the core recommendation problem grounded in an exploratory understanding of dataset distributions and domain constraints.
    - Top K & cold start
- **Prior Baseline & Critical Diagnosis:** Frame your starting point with reference to past project baselines (e.g., NCF, NMF), identifying their structural limitations, methodological gaps, or performance bottlenecks.
- **Group Contributions & Modeling Highlights:** Summarize your group's engineering, architectural, and analytical improvements over the baseline.
- **Evidence-Backed Quantitative Insights:** State key business and behavioral insights supported directly by concrete quantitative metrics, offline evaluation, or diagnostic error analysis

1. Data Understanding & Feature Engineering

Instead of repeating the interim report, focus on updating changes, data distributions, and incremental features developed (e.g., utilizing impression feedback, interaction types, or context signals). Use structured tables where appropriate.

- Changes: Remove slate_length + article_recency (Give explanation)
- Incremental features: Missingness Indicator
- Modify the tables

2. Models and Performance (Within 4 pages)

2.1 Architecture Diagrams

Provide clear, modular diagrams illustrating model architecture, input pipelines, and data flow

2.2 Experimental Setup

Report both training and testing performance across baseline and advanced models using appropriate ranking metrics with clear explanations of sample splits.

2.3 Performance Comparison

All recommendation models are evaluated using ranking-oriented metrics, including Area Under the ROC Curve (AUC), Mean Reciprocal Rank (MRR), Normalized Discounted Cumulative Gain at 5 (NDCG@5), and NDCG@10. These metrics assess the models' ability to distinguish clicked from unclicked articles and prioritize relevant news within the limited homepage Spotlight positions.

Model	AUC	MRR	NDCG@5	NDCG@10
UBCF + CBF	TBD	TBD	TBD	TBD
MLP	TBD	TBD	TBD	TBD
NRMS	0.6320	0.3386	0.3218	0.3851

NRMS Performance. The NRMS model is evaluated on the chronologically held-out Day 7 test set, comprising 73,152 impression records. To assess robustness against random initialization and negative sampling, the experiment is repeated using three random seeds (42, 43, and 44).

Random Seed	AUC	MRR	NDCG@5	NDCG@10
42	0.6369	0.3413	0.3256	0.3894
43	0.6273	0.3362	0.3186	0.3815
44	0.6317	0.3384	0.3211	0.3843
Mean ± Std	**0.6320 ± 0.0039**	**0.3386 ± 0.0021**	**0.3218 ± 0.0029**	**0.3851 ± 0.0033**

Table X: NRMS Test Performance Across Three Random Seeds

NRMS achieves a mean AUC of 0.6320 and NDCG@5 of 0.3218, demonstrating its ability to learn personalized ranking signals from news content and historical user interactions. The relatively small standard deviations across the three runs indicate consistent performance under the evaluated random seeds.

The model also exhibits improving validation performance during training. Validation AUC increases from 0.6193 in the first epoch to 0.6435 in the third epoch, while training loss decreases from 1.4516 to 1.3418. These observations suggest that the model successfully learns discriminative representations within the evaluated training configuration.

Nevertheless, aggregate ranking performance may conceal substantial differences across user segments. In particular, NRMS relies on historical clicked articles to construct user representations, potentially

limiting its effectiveness for users without interaction histories. This limitation is examined through a dedicated cold-start analysis and an architectural adaptation in Section 3.

2.4 Model-Specific Evaluation

Metric	Original NRMS	PG-NRMS	Relative Improvement
AUC	0.4875	**0.5631**	**+15.51%**
MRR	0.2582	**0.2848**	**+10.32%**
NDCG@5	0.2384	**0.2730**	**+14.50%**
NDCG@10	0.3004	**0.3329**	**+10.82%**

3. Contribution and Justification (1–3 pages per chosen aspect)

Complete the Self-Evaluation Table below. Groups are not required to cover all optional aspects; focus on the areas matching your team's technical depth.

<table>
<tr><th>Contribution Aspect</th><th>Dimension</th><th>Basic</th><th>Optional / Advanced</th></tr>
<tr><td rowspan="2">1. Data Understanding, Rich Signal Utilization & Feature Engineering (Impressions, multi-type user interactions, context, domain features)</td><td>Effort</td><td></td><td></td></tr>
<tr><td>Effectiveness</td><td></td><td></td></tr>
<tr><td colspan="4">Clarification:
Basic: e.g. Convert data into standard binary interaction pairs without deeper exploration of additional information; apply standard transformations or descriptive statistics.
Optional: e.g. Thoroughly analyze data distributions and sparsity prior to modeling; exploit rich signals (e.g., impression logs, exposed vs. unexposed items, multi-behavior interactions, timestamps/sessions) and engineer domain-informed features to enrich predictive signals.</td></tr>
<tr><td rowspan="2">2. Prior Baseline Benchmarking & Gap Diagnosis

(Critical evaluation of NCF, NMF, and conventional baselines)</td><td>Effort</td><td></td><td></td></tr>
<tr><td>Effectiveness</td><td></td><td></td></tr>
</table>

Clarification: ● **Basic:** e.g. Replicate standard baseline models (e.g., NCF, NMF) directly to obtain comparative metrics. ● **Optional:** e.g. Critically diagnose structural failure modes or information bottlenecks in baseline implementations; systematically justify your improvements against these identified gaps.			
3. Designing or Adapting Recommender Architectures	**Effort**		Implemented NRMS with multi-head self-attention and developed a popularity-gated user representation to address cold-start users without retraining.
	Effectiveness		Improved cold-start AUC by 15.51% and NDCG@5 by 14.50%, while preserving personalized recommendations for existing users.
Clarification: ● **Basic:** e.g. Direct implementation of standard models or simple ensembling. ● **Optional:** e.g. Adapt or customize architectures capturing temporal dynamics, sequential interaction patterns, or network/relational structures to improve recommendation quality.			
4. Model Diagnostics, Error Analysis & Quantitative Insights	**Effort**		Conducted multi-seed evaluation, self-attention ablation, and cold-start analysis to identify architectural limitations and assess ranking robustness.
	Effectiveness		Identified poor cold-start performance (AUC = 0.4875) and demonstrated improvements through popularity gating (AUC = 0.5631). NRMS achieved consistent results across three seeds (AUC = 0.6320 ± 0.0039).

<table>
<tr><td colspan="4">Clarification:
● Basic: e.g. Describe aggregate test metrics without deeper sub-group analysis
● Optional: e.g. Perform granular error diagnostics across distinct user/item segments; conduct deep learning model interpretation by examining learned parameters (e.g., analyzing attention weights, etc.); extract actionable behavioral and business insights supported by empirical evidence</td></tr>
<tr><td rowspan="2">5. Smart Use of AI Tools & Critical Collaboration</td><td>Effort</td><td></td><td></td></tr>
<tr><td>Effectiveness</td><td></td><td></td></tr>
<tr><td colspan="4">Clarification:
● Basic: Use AI tools strictly for routine coding syntax, formatting, or text polishing
● Optional: Engage AI as a critical sparring partner to explore modeling alternatives, catch conceptual/coding flaws, validate technical logic, and drive expanded learning documented in the AI Interaction Log.</td></tr>
</table>

4. AI Interaction Log (Appendix)

Provide a concise table documenting key collaborative interactions with AI tools:
1. Core Prompt & Task Objective: The specific question, modeling concept, or technical problem being explored.
2. AI Output Summary: The substantive response or code generated (do not include all of the code generated by AI in the Appendix).
3. Critical Evaluation: Your assessment of limitations, naive assumptions, or errors in the AI response.
4. Refinement & Final Action: How you validated, adjusted, and incorporated the solution into your project

5. References (Within 1 page)

Comprehensive citations for all datasets, baseline repositories, and referenced academic literature.